Probability and Statistics 1

Tutorial Sheet — Session 1

Basic statistical notions and vocabulary; frequency tables and graphs for a qualitative character

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Exercise 1

For each situation below, identify (i) the population, (ii) an individual, (iii) the sample, and (iv) whether the survey is a *census* or a *poll*.

(a) A university with 5000 students wants to estimate the average number of hours per week spent on programming. A questionnaire is sent to 200 students chosen at random.

(b) A security team analyses all 5000 login attempts recorded during one day on a server.

(c) A streaming platform asks 1000 randomly selected users, out of several million, what they think of a new recommendation feature.

Exercise 2

Classify each character below as *qualitative nominal*, *qualitative ordinal*, *quantitative discrete* or *quantitative continuous*. For each, give the set of modalities (or the set of possible values).

(a) Main programming language of a student.

(b) Number of failed login attempts on an account during a day.

(c) Severity of a vulnerability: low, medium, high, critical.

(d) Inference latency of a model, in milliseconds.

(e) Number of GPUs in a cluster node.

Exercise 3

A survey of 120 students asks for their preferred cloud provider. The answers are:

AWS 48, Azure 36, GCP 24, Other 12.

(a) Complete the frequency table below, giving the absolute frequency n_i, the relative frequency $f_i = n_i/n$ and the percentage $p_i = 100\, f_i$.

Cloud provider	n_i	f_i	p_i
AWS			
Azure			
GCP			
Other			
Total			

(b) Check that the relative frequencies sum to 1 and the percentages sum to 100.

(c) What is the mode of the distribution? Justify.

Exercise 4

Using the data of Exercise 3:

(a) Compute the angle $\theta_i = (n_i/n) \times 360^\circ$ of each sector of the pie chart.

(b) Verify that the four angles sum to 360°.

(c) Sketch the pie chart. Which provider occupies the largest sector? Which occupies the smallest?

Exercise 5

Using the data of Exercise 3:

(a) Draw a bar chart of the distribution. State clearly what the horizontal and vertical axes represent, and at which value the vertical axis starts.

(b) Draw a band diagram (horizontal bars) of the same distribution.

(c) In one or two sentences, compare the reading of a pie chart and of a bar chart for these data. Which one is more appropriate to compare the frequencies of Azure and GCP? Justify.

Exercise 6

A survey on the main code editor was conducted on $n = 200$ students. The following information is available:

- VS Code: absolute frequency $n_i = 95$;
- Jupyter Notebook: relative frequency $f_i = 0.225$;
- Vim/Neovim: percentage $p_i = 17.5\,\%$;
- emacs: absolute frequency $n_i = 25$.

(a) Complete the frequency table (absolute frequency, relative frequency, percentage).

(b) Check that the absolute frequencies sum to $n = 200$.

(c) Which modality is the mode?